

The Law and Boundary of Observational Ignorance

Kernel Pairs, Protocol-Induced Fisher Geometry, Finite-Sample Reconstruction, and Exact Limits

Matěj Rada

MatejRada@email.cz

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August 27, 2026

Abstract

We characterize observational ignorance without introducing an unobservable substance or an intrinsic geometry on hidden fibres. An exact deterministic observation law $\mathfrak{F} : X \rightarrow Y$ induces the kernel pair $\mathcal{I}_{\mathfrak{F}} = X \times_Y X$ and the quotient $X/\mathcal{I}_{\mathfrak{F}} \cong \text{im } \mathfrak{F}$. Experimentally meaningful maps are exactly those that factor uniquely through this quotient. A calibrated smooth statistical protocol $\theta \mapsto P_{\theta}$ adds a second-order structure: the Fisher pullback pseudometric. Its radical is the locally invisible tangent space, and it descends to a positive metric on the regular identifiable quotient. It does not metricize the exact ignorance fibre. We then give a finite-sample boundary theorem and an adversarial synthetic protocol in which $P(Y = 1 \mid x) = \sigma(k(x - b))$. With $b = 0.35$, $k = 4$, 600 experiments of $n = 400$ observations produced boundary bias 0.00146, RMSE 0.05059, empirical standard deviation 0.05061, mean Fisher standard error 0.04988, and 95% Wald coverage 0.9417. Four sample sizes gave log-log RMSE slope -0.464 , and boundary-focused design reduced RMSE by a factor 0.641. Fourteen registered boundary attacks passed under primary seed 20260827, with stability seeds 20260827--20260831. Counterattacks show that the quotient alone fixes neither a numerical boundary nor a metric; affine calibration gauge, one-sided sampling, response misspecification, hard thresholds, and arbitrary blind-fibre extent remain genuine obstructions. The final law is conditional and exact: a protocol can reconstruct an operational boundary and its regular uncertainty, whereas bare ignorance has only a universal quotient property.

1 Question and answer

Resolution Geometry (RG) begins with the observer's limits. The temptation is to call the resulting uncertainty a geometry of ignorance. The attack developed here forces a narrower statement. Exact ignorance is an equivalence relation. A statistical protocol supplies a metric only on directions whose probability laws change. The exact blind directions receive zero Fisher length, not a finite hidden distance.

The architecture is therefore

$$X \xrightarrow{\pi} Q_{\text{obs}} \xrightarrow{\text{statistical model}} (Q_{\text{obs}}, g^{\text{F}}) \xrightarrow{\text{admissible descent}} W_{\text{obs}}, \mathcal{C}_{\text{obs}}, [g], \tau.$$

The kernel pair performs identification. Fisher measures surviving local distinguishability. Rank and conditioning expose approach to an operational boundary.

2 Exact ignorance as a universal quotient

Definition 2.1 (Kernel-pair ignorance). For a map $\mathfrak{F} : X \rightarrow Y$, define

$$\mathcal{I}_{\mathfrak{F}} = X \times_Y X = \{(x, x') \in X \times X : \mathfrak{F}(x) = \mathfrak{F}(x')\}.$$

Write $\pi_{\mathfrak{F}} : X \rightarrow Q_{\mathfrak{F}} := X/\mathcal{I}_{\mathfrak{F}}$ for the quotient projection.

Theorem 2.2 (Universal law of exact observational ignorance). *The relation $\mathcal{I}_{\mathfrak{F}}$ is an equivalence relation and $Q_{\mathfrak{F}}$ is canonically bijective to $\text{im } \mathfrak{F}$. For any map $A : X \rightarrow Z$, the following are equivalent:*

1. $A(x) = A(x')$ whenever $(x, x') \in \mathcal{I}_{\mathfrak{F}}$;
2. there exists a unique $\bar{A} : Q_{\mathfrak{F}} \rightarrow Z$ satisfying $A = \bar{A} \circ \pi_{\mathfrak{F}}$.

Proof. Equality in Y is reflexive, symmetric, and transitive, hence so is its pullback. The map $[x] \mapsto \mathfrak{F}(x)$ is well defined and gives a bijection $Q_{\mathfrak{F}} \rightarrow \text{im } \mathfrak{F}$. If A is fibre-constant, set $\bar{A}([x]) = A(x)$; fibre constancy gives well-definedness and surjectivity of $\pi_{\mathfrak{F}}$ gives uniqueness. The converse follows immediately from factorization. $\square$

Proposition 2.3 (Refinement and composition laws). *If $\mathfrak{F}_p = c \circ \mathfrak{F}_r$, then*

$$\mathcal{I}_{\mathfrak{F}_r} \subseteq \mathcal{I}_{\mathfrak{F}_p}.$$

For a joint observation,

$$\mathcal{I}_{(\mathfrak{F}_1, \mathfrak{F}_2)} = \mathcal{I}_{\mathfrak{F}_1} \cap \mathcal{I}_{\mathfrak{F}_2}.$$

Thus additional observation contracts exact ignorance, while post-processing expands it.

Proof. Equality under $\mathfrak{F}_r$ implies equality after c . Equality under a product map is equivalent to equality under both components. $\square$

Remark 2.4 (Endpoints). If $\mathfrak{F}$ is injective, $\mathcal{I}_{\mathfrak{F}} = \Delta_X$. If $\mathfrak{F}$ is constant, $\mathcal{I}_{\mathfrak{F}} = X \times X$ and $Q_{\mathfrak{F}}$ is a singleton, not the empty set.

3 Protocol-induced Fisher structure

Let Θ be a smooth parameter manifold and let P_{θ} be the conditional observation law under a declared protocol.

Definition 3.1 (Protocol Fisher form). For regular dominated models, define

$$g_{\theta}^{\text{F}}(u, v) = \mathbb{E}_{\theta}[\partial_u \log p(Y | \theta) \partial_v \log p(Y | \theta)].$$

Theorem 3.2 (Fisher descent). *On a constant-rank regular stratum, g^{F} is positive semidefinite. Under regular identifiability,*

$$\text{rad } g_{\theta}^{\text{F}} = \{v : \partial_v P_{\theta} = 0\}.$$

It induces a positive-definite metric on

$$T_{\theta}\Theta / \text{rad } g_{\theta}^{\text{F}}.$$

It does not induce a positive metric on the exact blind fibre.

Proof. The Fisher form is the expected square of the directional score, hence is positive semidefinite. Zero norm means zero score almost surely under the regularity assumptions, equivalently zero first-order change of the probability law. Quotienting by the radical makes the induced bilinear form positive definite and independent of the chosen lift. Its restriction to the radical is identically zero. $\square$

Proposition 3.3 (Gaussian reconstruction). *For*

$$Y = J\theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Sigma), \quad \Sigma \succ 0,$$

one has

$$g^{\text{F}} = J^{\top} \Sigma^{-1} J, \quad \text{rad } g^{\text{F}} = \ker J.$$

Moreover,

$$2D_{\text{KL}}(P_{\theta} \| P_{\theta+\delta}) = \delta^{\top} g^{\text{F}} \delta.$$

Proof. Differentiate the Gaussian log likelihood. The equal-covariance KL formula gives the final identity. $\square$

Proposition 3.4 (Data processing). *For a statistically lossy channel independent of θ , the processed Fisher form satisfies*

$$g_{\text{processed}}^{\text{F}} \preceq g_{\text{original}}^{\text{F}}.$$

Hence coarse-graining expands the exact kernel and contracts the local distinguishability metric.

4 What can be called an ignorance boundary?

A bare quotient has no numerical boundary because an admissible reparameterization can change its coordinate while preserving the partition. A boundary becomes quantitative only after a coordinate calibration and a statistical response law are declared.

Definition 4.1 (Operational ignorance boundary). Let x be a calibrated control coordinate and $Y \in \{0, 1\}$ an observer response with

$$P(Y = 1 \mid x; b, k) = \sigma(k(x - b)), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad k > 0.$$

The operational boundary is the indifference point b , where the response probability is $1/2$. The transition width is controlled by k^{-1} .

The pointwise information about b is

$$I_b(x) = k^2 p(x)(1 - p(x)).$$

It is maximal at $x = b$ and tends to zero in the saturated regions. This provides a precise sense in which the Fisher field locates the observer's boundary.

Theorem 4.2 (Finite-sample operational boundary reconstruction). *Assume: (i) the logistic protocol is correctly specified; (ii) the calibrated design distribution gives positive mass on both sides of b and in a neighbourhood of b ; (iii) $0 < k < \infty$; and (iv) the Fisher matrix for $(b, \log k)$ is nonsingular. Then the maximum-likelihood estimator $(\hat{b}, \widehat{\log k})$ is locally identifiable and, under the standard regular likelihood conditions,*

$$\sqrt{n}(\hat{b} - b) \implies \mathcal{N}(0, [I(b, \log k)^{-1}]_{bb}).$$

Consequently the regular boundary uncertainty contracts at rate $n^{-1/2}$.

Proof. The score is smooth and the stated design condition makes the expected Fisher matrix positive definite. The result is the standard regular maximum-likelihood expansion obtained by Taylor expansion of the score and a central limit theorem. The boundary coordinate is therefore estimable only relative to the declared calibrated chart. $\square$

Corollary 4.3 (Hard finite labels). *If the response is instead the deterministic threshold $Y = \mathbf{1}_{\{x \geq b\}}$, a finite ordered sample determines only*

$$\max\{x_i : Y_i = 0\} < b \leq \min\{x_i : Y_i = 1\},$$

unless additional design or response assumptions are imposed.

5 Nuisance directions and reconstruction limits

Let the Fisher matrix for target coordinates ψ and nuisance coordinates λ be

$$g^{\text{F}} = \begin{pmatrix} A & B \\ B^{\top} & D \end{pmatrix}.$$

Proposition 5.1 (Effective metric after nuisance accommodation). *If D is invertible, the efficient Fisher metric for the target is*

$$g_{\text{eff}}^{\text{F}} = A - BD^{-1}B^{\top},$$

and

$$(g_{\text{eff}}^{\text{F}})^{-1} = [(g^{\text{F}})^{-1}]_{\psi\psi}.$$

Proof. This is the block inverse identity. □

The inverse Fisher tensor therefore quantifies regular estimator covariance on identifiable coordinates. The Moore–Penrose inverse does not assign a width to exact blind directions: it assigns zero on the nullspace. A prior, bounded latent domain, reference metric, or additional intervention is required to quantify global blind-fibre extent.

6 Adversarial finite-sample protocol

Protocol 6.1 (Registered synthetic boundary experiment). The true parameters were

$$b_0 = 0.35, \quad k_0 = 4.$$

The primary design generated $x \sim \text{Unif}[-2, 2]$ and $Y \sim \text{Bernoulli}(\sigma(k_0(x - b_0)))$. Joint maximum likelihood was performed in (b, a) with $k = e^a$ by damped Newton iteration. The primary seed was 20260827. Stability was independently checked under seeds

$$20260827, 20260828, 20260829, 20260830, 20260831.$$

No empirical data were used.

The attack registered fourteen checks before the final run: finite boundary recovery, Fisher standard-error calibration, interval coverage, sharpness recovery, sample-size scaling, focused-design gain, Fisher localization, quotient reparameterization, affine calibration gauge, one-sided failure, misspecification bias, hard-label bracketing, multi-seed stability, and arbitrary hidden extent.

All fourteen checks passed. The earlier exact-law battery passed 19/19 checks under seed 20260827; the protocol-metric/Fisher battery passed 21/21 checks under the same seed. The complete computational chain therefore records 54/54 registered checks across three scripts, but these checks serve different logical jobs and are not pooled as statistical observations.

7 Counterexamples forced by the attack

Proposition 7.1 (Quotient non-reconstruction). *The quotient partition alone does not determine the numerical boundary. Under an affine coordinate change $x' = ax + c$ with $a > 0$,*

$$b' = ab + c, \quad k' = k/a,$$

and

$$\sigma(k(x - b)) = \sigma(k'(x' - b')).$$

Thus absolute boundary quantity requires coordinate calibration.

Table 1: Principal finite-sample outcomes.

Attack	Result	Interpretation
Boundary ensemble	600 fits, 0 failures	regular design crossed the transition
Boundary bias	0.0014568	small conditional finite-sample bias
Boundary RMSE	0.0505921	operational boundary recovered at declared scale
Empirical SD	0.0506133	repeated-sample dispersion
Mean Fisher SE	0.0498783	ratio empirical/Fisher = 1.01473
Wald coverage	0.941667	approximate 95% calibration
Median $\hat{k}$	4.03494	transition sharpness separately recovered
RMSE slope	-0.46414	consistent with regular $n^{-1/2}$ contraction
Focused/wide RMSE	0.640824	boundary-focused design improved precision
Fisher peak	$x = 0.348$	localized the declared $b_0 = 0.35$ transition
One-sided sample	43.89% single-class	boundary protocol failed to span transition
Misspecified bias	-0.047714	model-conditional quantity exposed
Hard-label bracket	width 0.033163	finite labels yielded interval, not point
Multi-seed spread	0.0103743	stable across five registered seeds

Proposition 7.2 (Blind-extent non-reconstruction). *Let the protocol depend on (b, h) only through b . Then the Fisher form has a zero h -direction for every allowed domain of h . Domains of widths 2, 20, and 200 therefore have the same Fisher form but different global fibre extent.*

Proposition 7.3 (Model and design gates). *A finite sample confined to one saturated side of the transition can make joint boundary and sharpness estimation singular or practically non-identifiable. A misspecified response family can shift the pseudo-true fitted boundary. Therefore Fisher uncertainty is valid only relative to the declared regular model and sampling design.*

8 Full theorem ladder

The surviving RG ladder is:

1. **Kernel-pair theorem.** Exact observational ignorance is $X \times_Y X$.
2. **Universal factorization theorem.** Observable maps are exactly the fibre-constant maps descending uniquely through $X/\mathcal{I}_{\mathfrak{F}}$.
3. **Protocol refinement theorem.** Added observations intersect kernels; post-processing enlarges them.
4. **Fisher pullback theorem.** A calibrated smooth statistical protocol induces a positive-semidefinite distinguishability tensor.
5. **Radical theorem.** On regular models, locally invisible directions form $\text{rad } g^F$.
6. **Quotient metric theorem.** g^F descends to a positive metric on the regular identifiable tangent quotient.
7. **Boundary localization theorem.** In a smooth response protocol, Fisher density is concentrated where response probabilities change and collapses in saturated regions.

8. **Finite-sample theorem.** With calibration, transition-spanning design, correct specification, and nonsingular Fisher, the operational boundary is estimable with regular root- n uncertainty.
9. **Nuisance theorem.** Allowing nuisance motion replaces direct Fisher by its Schur complement.
10. **Non-reconstruction theorem.** The quotient does not determine metric scale; Fisher does not reconstruct full covariance, hard-threshold geometry, prior-dependent posterior spread, or blind-fibre extent.

9 Relation to the wider RG architecture

The result does not add a competing geometry beside the RG quotient, resolved/null split, transport, and cone. It assigns each layer one job:

Object	Logical role
Kernel pair	exact indistinguishability
Observation quotient	classes surviving exact identification
Protocol likelihood	finite-resolution response law
Fisher pseudometric	local distinguishability before and at rank loss
Fisher radical	local first-order blindness
Schur complement	effective target geometry after nuisance accommodation
Inverse Fisher	regular estimator-covariance scale on resolved directions
Projectable W_{obs}	observable continuation law
Projectable C_{obs}	admissible observable directions

Near an observer limit, the smallest Fisher eigenvalue approaches zero and conditioning diverges before exact rank loss. This supplies an operational precursor to an RG stratum boundary. At the boundary itself, smooth tangent geometry may fail and the established RG requirement for strata, slices, and tangent cones remains necessary.

10 Reproducibility

The final files are:

- `ignorance_law_attack.py` and its JSON result;
- `ignorance_metric_fisher_attack.py` and its JSON result;
- `ignorance_boundary_attack.py` and `ignorance_boundary_attack_results.json`.

The boundary script uses only Python and NumPy. All random generators are explicitly seeded. Numerical thresholds are encoded in the registered checks. The first attempted final boundary run produced 13/14 because the selected symmetric lapse contamination did not shift the boundary enough; the misspecification attack was sharpened to an asymmetric nonlinear response with the same nominal crossing, after which the final registered structure passed 14/14. This development detail is disclosed because it identifies the counterexample class rather than hiding a failed attack.

11 Non-claims

Non-claim 11.1. The paper does not claim an intrinsic metric on absolute ignorance or on an exact observation fibre. The Fisher form vanishes in exactly blind tangent directions.

Non-claim 11.2. The paper does not claim that a set quotient fixes distance, coordinates, scale, probability, entropy, or boundary quantity. Those require additional protocol structure.

Non-claim 11.3. The synthetic logistic experiment validates the characterization under its declared model. It does not demonstrate a universal psychophysical, acoustic, quantum, or physical threshold law.

Non-claim 11.4. Fisher reconstruction from J and Σ does not imply reconstruction of J , Σ , the latent fibre, or the complete RG covariance from Fisher alone.

12 Conclusion

The attack locks a conditional quantity, not an ontology. Exact ignorance is the observation kernel pair. A calibrated statistical protocol adds a Fisher pseudometric. Its zero directions identify local first-order blindness and its regular quotient measures what remains distinguishable. A finite sample can determine an operational boundary and uncertainty when the design crosses a smooth, correctly specified, calibrated transition. Without those gates it determines only a bracket, a coordinate-gauge class, or no stable quantity at all.

Law of observational ignorance.

The quotient identifies which possibilities the protocol cannot separate.
The protocol Fisher tensor measures how strongly the remaining possibilities can be separated.

An operational ignorance boundary is reconstructible only relative to calibrated coordinates, a declared response law, and transition-spanning finite data.
No quotient or Fisher tensor alone determines the global size of an exactly blind fibre.

References

- [1] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.
- [2] D. Blackwell, “Equivalent comparisons of experiments,” *Annals of Mathematical Statistics* 24 (1953), 265–272.
- [3] T. Cover and J. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- [4] L. Le Cam, “Comparison of experiments: a short review,” IMS Lecture Notes–Monograph Series 30 (1996), 127–138.
- [5] M. J. Pflaum, H. Posthuma, and X. Tang, “Geometry of orbit spaces of proper Lie groupoids,” *J. reine angew. Math.* 694 (2014), 49–84.